

# Web Application Vulnerability Scanner

## Introduction

Web applications are widely used for communication, business, education, and online services. However, these applications are often targeted by cyber attackers who exploit security weaknesses such as Cross-Site Scripting (XSS), SQL Injection (SQLi), and insecure HTTP configurations. The purpose of this project is to develop a Web Application Vulnerability Scanner that helps identify common security vulnerabilities and improves web application security awareness.

## Abstract

The Web Application Vulnerability Scanner is a Python-based security tool designed to perform basic vulnerability assessments on web applications. The system analyzes security headers, checks for potential XSS vulnerabilities, and identifies possible SQL Injection risks. The application also stores scan history, generates PDF security reports, exports scan data to CSV format, and provides dashboard analytics with graphical visualizations. This project demonstrates the practical implementation of cybersecurity concepts and secure web assessment techniques.

## Tools Used

- Python
- Flask
- SQLite3
- HTML
- CSS
- JavaScript
- Chart.js
- Pandas
- ReportLab
- Git & GitHub

## Steps Involved in Building the Project

1. Designed the user interface using HTML and CSS.
2. Developed the backend using Flask.
3. Implemented security header analysis functionality.
4. Added XSS vulnerability detection module.
5. Added SQL Injection detection module.

6. Integrated SQLite database for storing scan records.
7. Developed dashboard analytics with pie and bar charts.
8. Implemented scan history management and CSV export.
9. Generated professional PDF security reports using ReportLab.
10. Uploaded the project to GitHub with proper documentation and screenshots.

## **Conclusion**

The Web Application Vulnerability Scanner successfully identifies common web security weaknesses and provides detailed scan results through reports and dashboard analytics. The project enhanced practical knowledge of cybersecurity concepts, web application security testing, database integration, and report generation. It serves as a useful educational tool for understanding vulnerability assessment and secure web development practices.